Animal Welfare Enforcement in the UK: Evaluating Risk-based and Technology-led Approaches

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In 2024, 2.2% of UK farms were inspected for animal welfare compliance. This presents a major enforcement bottleneck where legal protections may fail to translate into real welfare improvements. This paper examines the factors that contribute to this enforcement problem and evaluates two potential solutions: a centrally-led risk-based approach to inspection prioritisation and the incorporation of precision livestock farming (PLF) technology into welfare monitoring. It concludes with proposals for implementation and further investigation.

The enforcement problem

The enforcement problem refers to the low rate of animal welfare inspections on farms across the United Kingdom. Currently, depending on the estimated number of farms, the average farm can expect an inspection once every 14 to 45 years (Animal Law Foundation, 2025). This paper focuses on on-farm inspections within England, Scotland and Wales, excluding other stages of the supply chain such as transportation and slaughter.

Responsibility for enforcement

In England, Scotland and Wales, responsibility for on-farm welfare enforcement is split between local and national authorities, with inspections typically conducted by local authorities and the Animal and Plant Health Agency (APHA) (APHA, 2016). While the Department for Environment, Food and Rural Affairs (DEFRA) in England and the Scottish and Welsh Governments set enforcement policy, they delegate much of its operationalisation to APHA, which in turn decentralises significant responsibility to local authorities. In practice, APHA often views its role as purely regulatory and advisory while local authorities remain the primary bodies for bringing prosecutions (Animal Law Foundation, 2025).

Enforcement triggers and process

Any farm can be inspected in England, Scotland and Wales, though there is no legal requirement specifying how often (APHA, 2016). In practice, checks are triggered by complaints, history or referrals rather than routine scheduling. Inspectors have power of entry and can act in the farmer's absence. Where unnecessary suffering is suspected, an inspection must follow within 24 hours and should normally be unannounced.

Enforcement in practice: key statistics

According to the Animal Law Foundation's 2025 report, 2.2% of UK farms were inspected in 2024, with 28.4% of those inspections finding non-compliance. Despite this, enforcement action is rare: 55.81% of local authorities in England, Scotland and Wales who identified breaches took no formal action and only 2.2% of non-compliance cases led to prosecution. Where action was taken, notices constituted 85% of all enforcement activity. In England, Scotland and Wales, 65.3% of farm-related complaints resulted in an inspection.

Whether a 2.2% inspection rate is unusually low compared to other regulatory services is unclear and perhaps beside the point. With approximately 28% of inspected farms found to be non-compliant, the

combination of low inspection coverage and high breach rates suggests that a significant amount of non-compliance goes undetected.

Scale and severity of potential suffering

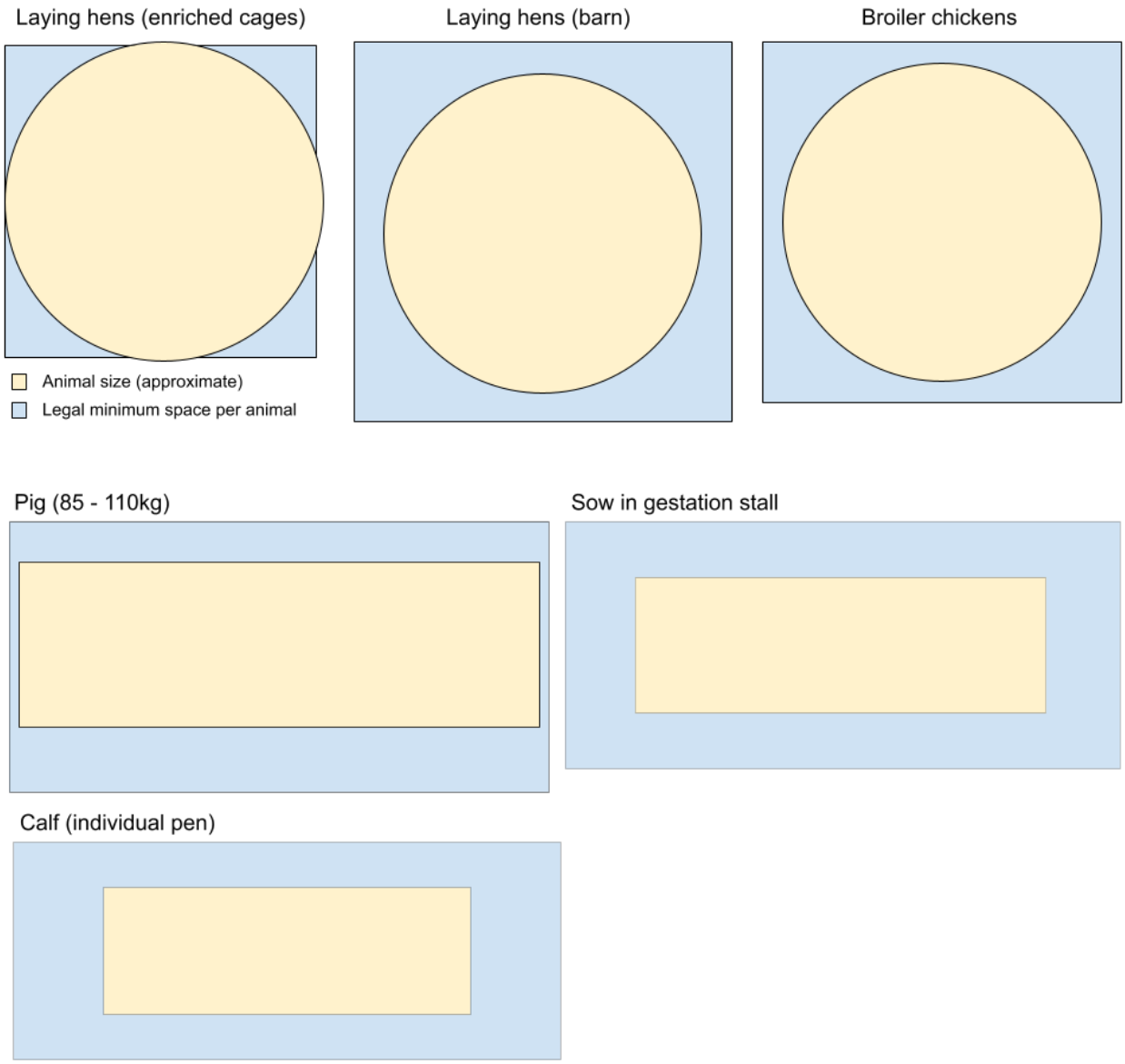
In order to explore the impact of undetected non-compliance, we need to consider the potential number of animals affected. Over one billion land animals were slaughtered in the UK in 2024/2025 (Food Standards Agency, 2025). Including farmed fish pushes the number of animals killed each year to well beyond one billion (Food Manufacture, 2025), and additional millions are estimated to die on farms before slaughter (The Guardian, 2023). Given that 28.4% of inspected farms are found to be non-compliant with animal welfare requirements, a substantial number of animals are therefore likely to be raised in conditions that risk falling short of legal standards, although the exact number cannot be precisely determined from available data.

It is also important to consider the potential severity of suffering. UK animal welfare regulation is primarily focused on preventing unnecessary suffering rather than promoting positive welfare. The baseline conditions and minimum space allowances below illustrate how low this legal threshold is in practice (Welfare of Farmed Animals (England) Regulations 2007).

- Animals must be inspected at least once a day (if welfare depends on frequent human attention) or at intervals sufficient to avoid suffering
- Animals that appear ill or injured must be cared for appropriately and without delay; if necessary, isolated with suitable accommodation and dry bedding
- Freedom of movement must not be restricted in a way that causes unnecessary suffering or injury
- Continuously restricted animals must get space appropriate to physiological and ethological needs, per good practice

Non-compliance therefore indicates that even these minimal protections, designed only to prevent the most basic forms of unnecessary suffering, were not being met.

Species	Minimum space per animal
Laying hens (enriched cages)	750cm2 total (599cm2 usable space)
Laying hens (non-cage/barn)	~1,111cm2 (9 hens/m2 max)
Laying hens (free-range)	~1,111cm2 indoor + 4m2 outdoor
Broilers (meat chickens)	1000cm2 at slaughter weight
Calves (individual pens, >220kg)	1.5 m2
Pigs (85 – 110kg)	0.65m2
Sows (gestation stall)	2.25m2



The role of assurance schemes

It could be argued that assurance schemes provide some regulation oversight. Assurance schemes were introduced in the UK in the 1990s to indicate to customers how food products were being produced (AHDB, 2025). In exchange for following specific welfare standards, farms are allowed to use scheme labels, such as Red Tractor or RSPCA Assured on their packaging. These schemes conduct their own inspections to ensure rules are being followed. If a farm is found to be non-compliant, it can be suspended or lose its membership entirely. Although comprehensive membership data is not publicly available, assurance schemes are a dominant force because major UK supermarkets require membership (e.g., Red Tractor) as a condition of sale (van der Ven et al., 2022). This has created a two-tier regulation system where the government sets the legal baseline and private assurance schemes act as more frequent auditors.

The existence of assurance schemes partly explains low government inspection rates: their presence creates an impression of oversight that can make the 2.2% inspection rate seem less alarming. However, many assurance schemes do not openly publish comprehensive inspection details including frequency and full results, which makes it difficult to assess their effectiveness (DEFRA, 2012).

Furthermore, there is a fundamental gap between the enforcement capacity of assurance schemes and the law. Assurance schemes are market-based, meaning they can strip a farm of a label or block it from selling its products in a supermarket but they cannot issue statutory notices, fines or bring criminal prosecutions. They cannot therefore be considered a substitute for state enforcement as they have no power to legally compel farms to comply with the law.

The lack of transparent data on membership, inspection frequency and outcomes make it difficult to assess the oversight assurance schemes provide. A 28.4% non-compliance rate among inspected farms nonetheless suggests existing oversight is insufficient. Given that state enforcement carries legal compulsion powers that assurance schemes lack, improving governmental enforcement seems to be the more reliable path to reducing non-compliance.

Key factors contributing to the enforcement problem

The problem of animal welfare regulation enforcement in the UK can partly be attributed to fragmented responsibility. Farm animal welfare enforcement is distributed across multiple authorities and shifts as animals move through the production chain. On-farm inspections are undertaken by local authorities as well as relevant government departments and agencies. With the exception of Northern Ireland, which operates a more centralised model, the enforcement structures are complex and involve multiple authorities (Animal Law Foundation, 2025).

In some cases, public bodies' stated responsibilities differ from their legal duties. For example, the Scottish and Welsh governments stated they did not enforce farmed animal welfare legislation, rather it is enforced by local authorities and in the case of the Scottish Government, APHA. Yet APHA stated they perform inspections only as a regulatory rather than enforcement body, and therefore they do not bring or refer prosecutions (Animal Law Foundation, 2025). This goes against APHA's regulatory and compliance policy that states it may refer matters for investigation or prosecution (APHA, 2017). The Scottish Government also claimed that APHA enforces farmed animal welfare legislation in Scotland, rather than the Scottish Government (Animal Law Foundation, 2025). These inconsistent accounts suggest a fragmented system where responsibility is unclear even to the authorities involved. Fragmented responsibility makes it difficult to monitor enforcement activity or identify where failures occur and creates conditions where authorities may defer responsibility to one another.

Resource constraints worsen the enforcement problem significantly. In Scotland there is one inspector for every 1327 farms; in England, one for every 892 and in Wales one for every 657. Northern Ireland with one inspector per 79 farms is an outlier, possibly reflecting its more centralised enforcement model. Across the UK as a whole, the ratio stands at one inspector per 490 farms (Animal Law Foundation, 2025).

Funding is likewise a significant constraint. A May 2025 report by the Chartered Trading Standards Institute found that 84% of local authorities considered cost a barrier to enforcement (Chartered Trading Standards Institute, 2025). Individual cases can exceed £200,000 when court delays, legal proceedings and the cost of caring for seized animals are factored in. More broadly, local authorities report insufficient

dedicated funding for animal welfare enforcement, covering staffing, legal capacity and physical infrastructure.

The current reactive approach lacks an effective deterrent because inspections are primarily triggered by complaints rather than routine scheduling. This sort of model allows certain farm types - such as new businesses, remote premises or indoor facilities - to avoid common inspection triggers. Consequently, welfare issues on these farms may remain undetected for significantly longer.

The slow pace of court proceedings further weakens the deterrent effect of the current system. Cited by 39.4% of surveyed officers as a key obstacle to enforcement (Chartered Trading Standards Institute, 2025), it can make prosecution seem too time-consuming an option for under-resourced local authorities.

Overall, fragmented responsibility, deficient resources, a reactive approach and slow court proceedings collectively reduce both the frequency of inspections and the likelihood that non-compliance, when found, results in meaningful enforcement action.

Institutional responses and policy developments

Following the release of Animal Law Foundation's reports, several bodies published independent evaluations of the enforcement system:

- February 2025: The Animal Sentience Committee issued a report noting that legislative compliance and enforcement were insufficient (Animal Sentience Committee, 2025)
- May 2025: The Chartered Trading Standards Institute identified financial costs and legal delays as primary barriers to effective enforcement (Chartered Trading Standards Institute, 2025)
- June 2025: findings were presented to the All-Party Parliamentary Group for Animal Welfare (APGAW, 2025)
- July 2025: A House of Lords scrutiny report raised concerns regarding non-compliance in the poultry sector and called for greater clarity on future enforcement guarantees (House of Lords Secondary Legislation Scrutiny Committee, 2025)

In December 2025, the government published its Animal Welfare Strategy for England (DEFRA, 2025). The document acknowledged the importance of enforcement in maintaining welfare standards; however, it did not propose new enforcement funding, structural reorganisation of regulatory bodies or changes to existing enforcement protocols.

The enforcement problem is not solely a question of resources - fragmented responsibility and unclear accountability suggest that additional funding alone would not resolve the underlying structural issues. The remainder of this paper evaluates two approaches, a centralised risk-based inspection framework and the incorporation of PLF technology into welfare monitoring; both aim to make better use of existing capacity rather than increase it. This is not because increased resources are unnecessary; the evidence presented above suggests they are. Rather, improved prioritisation is worth pursuing regardless of whether funding increases, because under-resourced enforcement that targets the right farms is preferable to under-resourced enforcement that does not. If resources do increase, then a well-designed risk-based model also ensures that additional capacity is deployed as effectively as possible.

Solution one: a centrally-led risk-based approach

What is a risk-based approach to the enforcement problem?

A risk-based approach to animal welfare regulation compliance is not novel. As mentioned previously, the government's current approach is indeed risk-based, with inspections being triggered in response to complaints (APHA, 2016). However, the approach described in this section aims to be preventative as opposed to reactive and ultimately involves targeting inspections towards sites where the likelihood of non-compliance is highest.

What risk-based approach should we aim for?

Considering the factors contributing to the enforcement problem, an effective approach should have several key criteria: it should improve detection efficiency (the rate of non-compliance detection per inspection) relative to the current complaint-led model, be easy for relevant authorities to implement, avoid perceptions of unfair targeting while supporting animal welfare objectives and maintain a baseline level of oversight to avoid certain farms becoming invisible to regulators.

How would the new approach work?

The goal of a new approach is to move from a complaint-led inspection schedule to a model that captures a wider array of non-compliance predictors. Such an approach would identify factors that statistically or logically predict a higher likelihood of non-compliance, allowing local authorities to prioritise limited resources.

One approach would involve analysing past inspection data to identify characteristics associated with breaches of animal welfare regulation. Analysts could apply statistical techniques such as regression analysis to determine which factors are most strongly linked to non-compliant outcomes and estimate the relative strength of these relationships. The result would be a model that uses historical enforcement data to identify patterns and predict which farms are more likely to breach welfare standards.

Alternatively, risk factors could be identified through expert judgement rather than statistical modelling. Senior inspectors and veterinary experts could review known indicators of non-compliance and assess their relative importance based on professional experience. For example, experts might conclude that a history of incomplete veterinary records is a stronger indicator of potential welfare breaches than factors such as farm size. This approach would rely on structured expert assessment to prioritise inspection targets without requiring a continuously updating statistical model.

Whichever approach is taken, the outcome should be that individual farms are assigned a risk score and ranked accordingly. Farms could then be grouped into three risk bands based on their predicted likelihood of non-compliance:

- High risk: farms within the highest predicted risk bracket (for example, the top 10 – 20% that inspection capacity allows). These could be inspected every 6 – 12 months, with unannounced visits to observe normal operating conditions.
- Medium risk: farms with elevated but not critical risk. Inspections every 12 – 24 months, using a mix of announced and unannounced visits.
- Low risk: farms with the lowest predicted risk. Inspections every 2 – 3 years while still maintaining oversight.

In terms of who might lead this approach, the Animal and Plant Health Agency (APHA) appears well placed. APHA already collects data on welfare complaints, inspection activity and non-compliance findings (House

of Commons, 2025), making it a strong candidate for building and maintaining such a dataset. It has also identified data science capability in its 2021 - 2026 science strategy (APHA, 2021). Under this approach, local authorities could feed enforcement activity back to APHA through a national reporting system. As a central body working with local authorities and employing veterinary experts, APHA could coordinate both the data analysis and incorporation of expert judgement.

Safeguards

To prevent farms from becoming effectively invisible:

- Random inspections should be conducted across all bands to test the accuracy of the risk assessment and help identify additional factors that may predict non-compliance, rather than only reinforcing those already used to prioritise inspections.
- Complaints should trigger automatic assessment and, where credible, an inspection that overrides the ranked list
- Inspection capacity should be allocated in the following order:
    - Complaints trigger an automatic response
    - Remaining capacity is allocated according to the ranked risk list
    - Any deviation from the ranked list must be documented with reasons

This ensures the risk assessment supports rather than replaces professional judgement and complaint-led enforcement.

How would it realistically be introduced?

A pilot programme could be introduced across a limited number of local authorities with varying geography and farm density. The first step would be to develop the risk assessment using enforcement and farm data, then produce risk bands and determine the inspection procedure for each band (for example, inspection frequency and whether visits are announced or unannounced). The selected local authorities would then use these risk scores to plan inspections. Inspection outcomes would be recorded and fed back to the central analytical team to refine the risk assessment. Regular meetings between local authorities and the central team would help address questions, flag issues and resolve problems as they arise.

How will we know if the approach is successful?

1) Detection rate: if the new approach finds a higher rate of non-compliance per inspection than the current complaint-led approach, this suggests the targeting is working. A rising non-compliance rate among inspected farms should not be misread as standards declining - it is the expected and desirable outcome of concentrating inspections where breaches are most likely. Comparing the pilot area to a non-pilot control area helps confirm this: if the non-compliance rate rises in the pilot zone while remaining stable elsewhere, it indicates the model is successfully identifying farms where breaches were previously going undetected.
2) Sustainable workload: the number of inspections per officer should remain stable or decrease. If the new model overloads inspectors with too many “high-risk” flags, the system will fail regardless of how accurate it is. Success means the model is operationally realistic and manageable for the inspectors.

3) Efficiency (cost per breach): we should see the cost of finding each breach go down. If the system is working, the money spent on inspections should be producing more results. By targeting the farms most likely to fail, the government gets better "value for money" out of its enforcement budget.
4) Predictive accuracy: we will know the risk assessment is reliable if the farms labelled "high-risk" actually have more breaches than those labelled "low-risk". If the high-risk farms turn out to be mostly compliant, the risk assessment is targeting the wrong factors and should not be scaled up.

What failures should be anticipated?

- Inaccurate risk assessment/model: if the available data for a model are insufficient or poor quality, or the expert-led approach misses key factors, the risk assessment may fail to predict non-compliance reliably. In that case, farm categorisation and the resulting inspection priorities are unlikely to improve detection.
- Implementation cost: the new approach would require additional resources, such as a central analytical or coordination team and increased costs associated with handling detected breaches. In the short term, improved targeting may uncover more non-compliance, increasing enforcement costs. But in the longer term, more effective detection may act as a deterrent and reduce the number of breaches thereby decreasing enforcement costs.
- Fragmented responsibility: since fragmented responsibility is a clear contributing factor to the enforcement problem, the approach will only be effective if roles and responsibilities are clearly defined. Without clear coordination between authorities – or clarity within authorities about who is responsible for animal welfare inspections – the approach risks reproducing existing governance problems.
- Slow court proceedings: the approach does not address delays in prosecution. As with implementation costs, improved detection may initially add pressure to an already slow court system. Over time, however, stronger detection and enforcement may reduce the number of cases as non-compliance becomes less attractive.

Does this solution improve on the current fragmented system?

As mentioned previously, a fragmented system is a major contributor to the enforcement problem. Therefore, a system which increases consistency between local authorities (clear, transparent and standardised approach which is easier to monitor and make improvements to) is preferable.

This approach should reduce inconsistency in enforcement across local authorities since risk scores will be calculated for all farms using the same standardised risk assessment and each authority will need to prioritise inspections based on this.

Under this proposed approach, the roles and responsibility of a central regulator and local authorities must be clearly defined. The central regulator would be responsible for developing the risk assessment and regularly reviewing available data and evidence to identify non-compliance risk factors. They would then return this information to local authorities with the requirement that farms are inspected in order of highest risk first.

Local authorities would be responsible for recording and reporting enforcement activity relating to farms in their area, including complaint and prosecution history. Once they receive the risk list, they would undertake inspections in order of priority, beginning with the highest-ranked farms first.

The central regulator would still retain the ability to conduct inspections themselves, but primary responsibility would remain with local authorities.

Does it meaningfully increase welfare protection under limited resources?

Ultimately, improved detection efficiency would mean identifying more non-compliance within existing inspection capacity. This depends on whether centralised risk assessment outperforms the current model in which local authorities rely mostly on local intelligence and complaints to plan inspections.

We can learn how to best implement the new approach by looking at previous risk-based approaches introduced by the government. For example, the Food Standards Agency (FSA) pilot programme (January 2021 – March 2022) aimed to improve food safety compliance by targeting areas posing the greatest risk (FSA, 2023). It reviewed the existing food safety framework and introduced tools such as a business risk scoring system and decision matrix to form the Food Standards Delivery Model (FSDM). Seven local authorities were trained to implement the approach, and they reported that the FSDM scheme assessed the risk profile of food businesses more accurately and the model integrated relatively easily into existing practices. A key strength of this pilot was the strong communication, training and ongoing support between central government and local authorities.

We can also learn from approaches that have been criticised. The post-Brexit Border Target Operating Model has been challenged over concerns that its new risk categories were too broad (House of Commons Environment, Food and Rural Affairs Committee, 2025). Instead of creating more precise targeting, critics argue it simply increased the number of checks across many different groups. This serves as a warning for animal welfare reform: the goal must be smarter targeting and a more predictable inspection schedule, rather than just casting a wider net that increases the burden on everyone without a clear benefit.

On the whole, this approach should start as a pilot built on close, frequent communication between a central regulator and local authorities. This allows for early troubleshooting and ensures that inspection rules only change when evidence shows they need to. By staying focused on genuine risks, the system avoids the mistake of simply creating more work for everyone without improving animal welfare.

Could farms be misclassified?

There is a danger that high-risk farms are labelled low-risk. This may mean that non-compliance goes undetected as inspection frequency is low. It is possible that this will happen in the short term, but as the model gathers more data on farms, in the long run the hope is it will output a more accurate prediction of what factors are associated with non-compliance. Overall, the model should be accurate enough that high-risk farms are more likely to be non-compliant than low-risk farms and that serious welfare harm does not routinely occur in farms classified as low-risk.

Could it entrench low inspection capacity?

Optimisation could reduce the need for increased funding by demonstrating improved efficiency. However, improved efficiency does not necessarily mean sufficient welfare protection. Therefore, its success should be evaluated not just by detection rates, but by whether overall welfare outcomes improve

meaningfully; this could be indicated by the non-compliance rate falling over time and high-risk farms improving after targeted inspections. If optimisation still leaves substantial under-enforcement, it strengthens the case for increased governmental resources.

Could it distort behaviour?

There is a risk that farms could "play the system": avoid displaying indicators that predict non-compliance and yet still not be compliant with animal welfare regulation. Or if farms learn of their risk band, especially low-risk farms, this would lower the incentive to remain compliant based on the belief that inspection frequency would be low. This risk is mitigated though not eliminated by the inclusion of random inspections across all bands and complaint-triggered overrides. However, the deterrent effect depends on the proportion of random checks and whether risk bands are disclosed to farms.

Overall assessment

I think this approach is the best risk-based option given limited resources: it provides a clear and transparent way of prioritising inspections that, in theory, should improve the detection of non-compliance. However, its success would depend on strong and frequent communication between local authorities and central regulators as well as an easy-to-use data sharing platform. The approach should also remain flexible in order to address urgent welfare issues, prioritising real-time complaints first, followed by the risk list and then other forms of local intelligence.

A pilot programme should initially be launched to test feasibility and gather feedback on how best to implement the system. The early stages of implementation will likely require the most work - establishing processes and addressing key issues early. Over time, the risk assessment could be refined and improved as more information becomes available. If overall welfare outcomes do not improve, even assuming the system operates as effectively as possible, this would suggest that government resources allocated to enforcement may need to increase.

Solution two: PLF as continuous monitoring

PLF welfare monitoring

If precision livestock farming (PLF) technology becomes more widely adopted on farms, and if the data it generates can be used to assess animal welfare, there may be an opportunity to incorporate it into regulatory oversight. This section examines whether both conditions hold and what it would mean in practice.

What is PLF?

PLF uses tools like sensors, cameras and microphones to monitor animals and their environment. These devices collect data on movement, sound, temperature and location which a system then uses to trigger a response. This could be an automated action, such as refilling a feeder when the level is detected to drop, or an alert sent to a farmer, such as a notification that an animal has wandered outside its boundary or is showing early signs of lameness (Carolin et al, 2017).

The primary goal of PLF is to maintain individual animal oversight as farm sizes increase. On large-scale farms, it is improbable for human staff to monitor every animal at all times; PLF bridges this gap by providing continuous, individual-level attention. Beyond welfare, these tools improve efficiency by

automating routine tasks adjusting lighting and feeding and can boost productivity by identifying the specific factors that lead to the greatest yields (Bernabucci, G. et al., 2025).

Current adoption trajectory

There are no up-to-date official statistics indicating the overall rate of PLF adoption in the UK, but 2019 survey data from England shows that uptake is technology-specific and uneven across sectors (DEFRA, 2020). For example, while 64% of pig and poultry farms use frequent automated weighing to monitor growth, fewer than 10% of grazing livestock farms use pasture sensors.

The primary driver for adoption is economic: 78% of surveyed farms cited increased productivity and performance as their motivation, while only half identified animal health and welfare improves as a factor. Significant barriers remain, including the perceived cost-effectiveness of the tools and their relevance to specific farming types (DEFRA, 2020).

To encourage innovation, the government provides support through broader schemes such as the Farming Equipment and Technology Fund and the Farming Innovation Programme (DEFRA, 2026). While these are not exclusive to PLF, they provide the financial pathways for technological investment. AS PLF tools become more affordable and their productivity benefits more widely demonstrated, adoption across the industry would be expected to increase, particularly given the sector's broader shift towards reducing reliance on manual labour.

Could PLF farming technology be used to assess animal welfare?

This table shows the types of technologies used to measure different welfare indicators related to pig production (Gómez et al., 2021).

Welfare indicator	Inputs	Measured traits
Activity & moving	Cameras, accelerometers (wearables), thermal cameras, RFID*	General motion (active/inactive state), tracking location, postural states (standing and lying), and transition between states.
Feeding & drinking	RFID, cameras, accelerometers, load cells, microphones	Intake volume, frequency and duration of feeding/drinking; nursing/suckling behaviour or social competition at stations.
Social & specific behaviours	Cameras, accelerometers, microphones	Aggression, social grouping and species-specific stress behaviours (e.g., tail-biting, mounting)
Physical condition	Load cells, 3D cameras, microphones, thermal cameras	Gait and weight distribution (lameness detection), body weight/condition scoring, vocalisations (coughs/distress) and surface body temperature
Health status	Cameras, accelerometers, infrared thermography, RFID, microphones	Early detection of disease (e.g., respiratory issues, diarrhoea or mastitis), hoof lesions and thermal

		changes indicating fever or inflammation.

* Radio frequency identification: wireless technology that identifies, tracks and monitors individual animals in real time.

Sensor to action process

1) Collection: sensors capture raw signals from the animals or their environment.
2) Analysis: software filters this raw data to identify specific patterns, such as the sound of a cough or a change in walking speed.
3) Output and action: the processed data may result in a change on the farm. This may be automatic such as a system adjusting ventilation or light levels in response to sensor data or alert-based where a dashboard notifies the farmer of a red flag that requires a manual human response (Kleen and Guatteo, 2023).

How PLF can be technically used to ensure compliance with animal welfare regulation

From a technical perspective, PLF can be used to gather, process and interpret data on animal welfare in farm settings. One study determined there are over 80 PLF tools commercially available that can potentially be used to assess animal welfare indicators on pig farms (Gómez et al., 2021). PLF tools can target specific areas of welfare regulation. For example, overhead cameras can be used to measure pen occupancy to determine whether stocking density regulation is adhered to. Alternatively, PLF can be used to verify the UK law banning the confinement of calves over eight weeks old to individual stalls and pens – accelerometers can detect play and grooming between calves. Cameras can also be used to check for illegal handling of animals (Gómez et al., 2021)..

However, the existence of these technologies does not automatically determine how PLF data should be incorporated into regulatory systems and what type of systems would be best suited to PLF integration. Several design considerations would need to be addressed around procedural fairness, uneven applicability across farms and data management for PLF data to effectively support enforcement.

To ensure compliance with UK standards of procedural fairness, PLF data should function as a regulatory trigger rather than a prosecution tool. Rather than outputting a welfare score - which raises black box concerns - systems should output a summary of raw facts (e.g., animal #402 spent 40% less time standing than the previous seven-day average) flagging where inspections should be undertaken. The inspector would retain expert witness status while leveraging the continuous monitoring capacity of PLF technology. This complementary role is also important given that PLF is harder to implement on extensive grazing farms (sheep and cattle) than in intensive indoor settings (chickens and pigs), meaning it cannot easily be applied uniformly across the sector and should not replace existing inspection efforts (Morgan-Davies et al., 2017). A practical further consideration is data volume: rather than a constant stream, periodic summary reports would be more manageable for inspectors, preserving the benefits of continuous monitoring without overwhelming existing workflows.

Governance models

It seems plausible that PLF inputs can technically grab meaningful data related to animal welfare. But how is this data managed and who should see it? There are four key governance models when considering the utilisation of PLF as enforcement of animal welfare regulation.

The first is the farmer-managed feedback loop: PLF systems are chosen by farms and data is fed back to farmers who act upon welfare alerts. This would offer the lowest level of transparency as only farmers would know if welfare is being handled or not. It would be easiest in terms of implementation as it only requires one level of welfare handling rather than sharing the role of regulation with multiple groups. However, enforcement capacity would be low: whether a welfare alert is acted upon is entirely up to the discretion of the farmer; there would be no external incentive to act. Whilst this governance model is the easiest to implement and least controversial, it is the least likely to ensure compliance. Whilst the farmer-managed approach is the easiest to implement (is most likely the default path) and least objectionable by farmers, it offers the weakest guarantee of compliance enforcement and transparency and therefore is not the best governance model.

The second model is company integration, where welfare data is shared with the PLF provider, who monitors and enforces compliance. Implementation would be relatively easy, since companies already manage the data infrastructure. Enforcement would be set by individual companies and could range from financial penalties to softer incentives like benchmarking or gamification. Under this approach, companies would have weak commercial incentives to impose strict welfare safeguards, especially if these safeguards make products less appealing. Transparency remains limited and enforcement would likely vary between providers. Overall, this model offers more oversight than the farmer-managed approach, but it still relies on private companies to uphold legal standards. At best it could complement existing regulation, but it is unlikely to ensure strong or consistent compliance on its own.

The third option is farm assurance scheme integration. Under this model, farms would be required to share relevant PLF data directly with the assurance schemes they belong to. This creates a data-led audit system where schemes can periodically review digital records alongside physical inspections. In practice, this would require schemes to accredit specific PLF systems that provide welfare-relevant data and demonstrate high accuracy. To incentivise farmers to share system-generated data, they could be rewarded with “earned recognition”, resulting in fewer physical inspections (FSA, 2024). In theory, this governance model is one of the strongest for accountability as the stated intention of schemes is to uphold welfare standards. However, risks remain: if schemes both approve PLF systems and judge compliance using the data those systems generate, conflicts of interest may arise. There is also a risk that accreditation decisions could be influenced by commercial partnerships between tech companies and schemes rather than being drive purely by animal welfare outcomes. Overall, using PLF data in assurance schemes helps reward compliant farms with fewer inspections while flagging those that need attention. However, it seems that government-created safeguards are required to ensure that animal welfare is not sacrificed for profit.

The final model is direct government integration. Under this approach, any PLF system collecting welfare-relevant data would be required to make that information accessible for regulatory oversight. For example, PLF providers could enable access for central regulators who would audit the information and delegate enforcement to local authorities. Implementation would require legislation mandating government access to the records held by PLF companies, alongside a standardisation of welfare metrics. These standardised reports could be generated by the tech companies or produced by regulators from

raw data. However, direct company-to-government transmission may raise serious concerns for farmers regarding surveillance and data ownership. A more workable approach might be to require PLF systems to generate a standardised welfare report that goes to the farmer first, who then submits it as part of their legal compliance. There is also a practical risk: if government oversight is perceived as punitive rather than supportive, farmers may be discouraged from adopting PLF systems that generate welfare data. Overall, poor implementation of this approach could simply create resistance without improving outcomes for farmed animals. Therefore, for this model to succeed, it must be carefully designed to ensure credibility, fairness and genuine welfare improvement

Comparative assessment: assurance scheme vs government integration

Enforcement and institutional integrity

Government integration provides more direct and legally grounded enforcement because those responsible for interpreting compliance are those with the authority to sanction breaches. Contrastingly, assurance schemes introduce an intermediary layer whose effectiveness must itself be monitored.

Assurance schemes face possible conflicts of interest, especially if they both approve PLF systems and evaluate compliance. Government enforcement is likewise subject to pressure from farmers, PLF companies and other relevant parties, although it is more institutionally independent.

Scale and severity

Not all farms are members of assurance schemes, whereas all farms are subject to government enforcement. Government integration would therefore be better at ensuring that a greater number of farms meet at least the minimum animal welfare standards set out in law.

It is possible that government integration could bring animals both up and down to basic welfare baseline. Assurance schemes may be better positioned to secure higher levels of welfare: whilst some schemes begin with the legal minimum as a baseline, some, such as RSPCA Assured and organic standards, go beyond statutory requirements (Bartlett et al., 2023). Schemes that integrate PLF could therefore embed welfare standards that exceed the legal minimum rather than focusing solely on baseline compliance.

Feasibility and political durability

Assurance schemes may allow for quicker implementation since some schemes already have membership portals and auditing processes in place (Red Tractor, 2025). Adapting these existing systems to include PLF data is a smaller step than starting from scratch. In contrast, government integration would likely require significant legislative reform and a much large administrative capacity to build and manage a national data network.

Both forms of integration must demonstrate acceptability and longevity. A model is only feasible if it avoids significant backlash from farmers, the public and the industry; an optimised welfare solution is of little value if political pressure leads to it being scrapped. To be successful, a model must have longevity and be designed to survive political and industry shifts. While this is a challenge for both approaches, assurance schemes are arguably more viable because they are less bureaucratic: their smaller scale allows them to adapt quickly, troubleshoot problems and respond to industry feedback more effectively than a government department.

Final judgement

In the short term, assurance schemes are best placed to pilot PLF data as welfare indicators. Because they are already equipped to manage audits and labels, they can iterate and solve problems more quickly than governmental departments.

In the long term, if assurance scheme pilots prove successful, the government could move towards full integration. This would require farms with PLF systems to submit key welfare statistics directly to the regulator. This participation could be incentivised through earned recognition (rewarding farms with fewer physical inspections unless the data reveals patterns of non-compliance). Ultimately, government integration remains the long-term ideal due to its stronger enforcement capability and the fact that it can observe farms that are not part of assurance schemes.

Discussion

Responsibility for farmed animal welfare regulation in Great Britain is effectively shared between APHA and local authorities, with inspections often triggered by complaints. Much of the ongoing oversight appears to rest with assurance schemes, where non-compliance risks loss of membership rather than legal prosecution. While APHA encouraged authorities to prioritise local intelligence over centrally set inspection schedules (DEFRA, 2022), this reactive approach has not improved outcomes: as the Animal Law Foundation (2025) reports, inspection and non-compliance rates remain stagnant, leaving the system unclear and disjointed.

Rather than distributing responsibility across local authorities without clear central coordination, it seems that a government-led enforcement strategy would improve transparency (centralised access to data across all farms), consistency (application of statutory standards rather than varying private ones) and legal accountability (where non-compliance carries statutory consequences rather than membership sanctions alone).

A truly effective risk-based model would require a designated central regulator to manage a standardised risk-assessment framework (using either statistical modelling or expert analysis). It would also require unified data-sharing infrastructure where local authorities submit intelligence for central evaluation and regulators then incorporate this data into their risk models. Roles would also need to be clearly delineated, with central regulators determining farm risk levels and communicating these to local authorities who retain responsibility for on-the-ground enforcement. Before national rollout, a geographically limited pilot should be launched across a mixture of local authorities to test feasibility and gather inspector feedback.

It should be noted that a risk-based model is an efficiency reform rather than a structural one. Its value depends on how much targeting inefficiency exists in the current system. If meaningful welfare outcomes do not improve under this model, this would indicate that inspection capacity itself rather than its allocation is the key constraint and would strengthen the case for increased resourcing.

In terms of PLF as continuous monitoring, integration with assurance schemes may be more feasible in the short term. Incorporating PLF systems into statutory regulation would likely require coordination across multiple government departments as well as the development of new legislation and regulatory frameworks. In contrast, private assurance schemes can typically update their standards more rapidly. Because PLF-based welfare monitoring is still developing, these schemes may provide a more suitable

environment for early experimentation and refinement before the technology is incorporated into statutory enforcement.

Meanwhile, the government could introduce legislation around PLF. Firstly, the government should regulate which PLF products are allowed on the market. Every system must be tested in commercial farm conditions to prove it does not harm animals. For example, a tool designed to speed up production should be banned if it interferes with an animal's basic wellbeing or violates welfare laws. Secondly, assurance schemes must be transparent about how to interpret raw data as welfare indicators. By making their methods public, the government and the general public can check and challenge whether these schemes are actually protecting animals of simply "humane washing" the data. Finally, there should be rules to prevent assurance schemes from picking tech based solely on how lucrative it is for the farm. Any system they approve must clearly prove that it improves the lives of animals, not just the efficiency of the business.

Conclusion

The enforcement problem has structural issues that additional resources alone would not resolve. Fragmented responsibility, a reactive inspection model and limited transparency have produced a system where basic animal welfare protections go largely unenforced.

A centrally-led risk-based approach offers an actionable path forward, directing existing inspection capacity where it is more needed, though its gains will be limited if inspection capacity itself remains the binding constraint. In the longer term, PLF technology presents an opportunity to move beyond the snapshot inspection model, but only if its data is managed transparently and in the interests of animal welfare rather than commercial efficiency.

Neither solution is a quick fix and both depend on a more coherent institutional framework than currently exists. The question is not whether better enforcement is achievable, but whether there is sufficient political will to reform a system that has remained stagnant for too long.

Bibliography

Agriculture and Horticulture Development Board (2025) UK Farm Assurance Review

All-Party Parliamentary Group on Animal Welfare (2025) APGAW Roundtable on Enforcement

Animal and Plant Health Agency (2016) Animal welfare on farms inspection

Animal and Plant Health Agency (2017) Regulatory and Compliance Policy

Animal and Plant Health Agency (2021) APHA Science Strategy 2021 to 2026

Animal Law Foundation (2025), The Enforcement Problem: 2024 Data Report

Animal Sentience Committee (2025) Report on the Due Regard to Animal Welfare: Legislative Compliance and Enforcement

Bartlett, H., Balmford, A., Holmes, M.A. and Wood, J.L.N. (2023) Advancing the Quantitative Characterization of Farm Animal Welfare, Proceedings of the Royal Society B: Biological Sciences

Bernabucci, G. et al. (2025) Precision livestock farming: an overview on the application in extensive systems, Italian Journal of Animal Science

Carolin, J., Yasotha, A. and Sivakumar, T., 2017. Precision livestock farming: An overview. Indian Journal of Animal Production and Management

Chartered Trading Standards Institute (2025) Tackling Animal Harm

Department for Environment, Food & Rural Affairs (2020) Farm Practices Survey Autumn 2019: England

Department for Environment, Food & Rural Affairs (2022) Animal Health and Welfare Framework

Department for Environment, Food & Rural Affairs (2025) Animal Welfare Strategy for England

Department for Environment, Food & Rural Affairs (2026) Farming Equipment and Technology Fund 2026: Guidance Now Available, Defra Farming Blog

Department for Environment, Food and Rural Affairs (2012) Farm Animal Welfare: Health and Disease

Department for Environment, Food and Rural Affairs and Animal and Plant Health Agency (2017) Animal Welfare Inspections on Farm

Food Manufacture (2025) Farmed fish welfare put on the agenda in Scotland, Food Manufacture

Food Standards Agency (2023) Evaluation of Food Standards Pilot: Introduction

Food Standards Agency (2024) Earned Recognition: Approved Assurance Schemes.

Food Standards Agency (2025) Animal Welfare Report 2024/25

Gómez Y, Stygar AH, Boumans IJMM, Bokkers EAM, Pedersen LJ, Niemi JK, Pastell M, Manteca X and Llonch P (2021) A Systematic Review on Validated Precision Livestock Farming Technologies for Pig Production and Its Potential to Assess Animal Welfare

House of Commons (2025) Written Question: Animal Welfare: Crime (UIN 73917)

House of Commons Environment, Food and Rural Affairs Committee (2025) UK-EU Trade: Towards a Resilient Border Strategy. Fourth Report of Session 2024-25

House of Lords Secondary Legislation Scrutiny Committee (2025) Thirty-Second Report of Session 2024-25

Kleen, J.L. and Guatteo, R. (2023) Precision Livestock Farming: What Does It Contain and What Are the Perspectives?, Animals

Morgan-Davies, C., Lambe, N., Wishart, H. and Waterhouse, A. (2017) Impacts of Using a Precision Livestock System Targeted Approach in Mountain Sheep Flocks, Livestock Science

Red Tractor (2025) New Portal Features: Making Life on the Farm Easier, Red Tractor

The Guardian (2023) Premature death of 80m chickens raises concerns over UK's fast-growing breeds, The Guardian

The Welfare of Farmed Animals (England) Regulations 2007

van der Ven, C. et al (2022) Core Environmental Standards for UK Imported Agri-Food Products: Options for Pesticide and Fertiliser Use